

Mirror & water images

COMPLETE LEARNING GUIDE

What You'll Learn

Mirror image and water image questions test whether a candidate can correctly apply two distinct, mechanical geometric transformations — a left-right flip (mirror, as in a vertical mirror placed beside an object) and a top-bottom flip (water image, as in a reflection in a pool below an object) — to letters, numbers, clocks, and calendar dates. The two transformations are frequently confused with each other because both are 'reflections,' but they flip along different axes and therefore produce different results on the same input, especially for asymmetric characters and clock times. This sub-module covers the mechanics of each transformation, dedicated formulas for mirror and water images of clock times (a very common question type), how digits and letters behave differently under each flip, symmetry-axis identification, and combined two-step problems.

1. Mirror Image Basics: Left-Right Reversal

A mirror placed vertically beside an object reverses left and right: the order of characters in a word reverses (last character appears first), and each individual character's shape also flips horizontally — characters with a vertical line of symmetry (like A, H, I, M, O, T, U, V, W, X, Y) look unchanged, while others (like B, C, D, E, F, G, J, K, L, N, P, Q, R, S, Z) look visibly different or unrecognizable.

- Mirror image of 'MATHS': since M, A, T, H are all vertically symmetric letters, and reading order reverses, the mirror image reads as the letters in reverse order (S, H, T, A, M), with S also flipping shape since it isn't vertically symmetric.

2. Water Image Basics: Top-Bottom Inversion

A water (or 'reflection in a pool') image flips an object top-to-bottom instead of left-to-right — the left-right ORDER of characters stays the same, but each character's shape flips vertically, so only characters with a horizontal line of symmetry (like B, C, D, E, H, I, K, O, X) look unchanged.

3. Mirror Image of Clock Times (Formula-Based)

The mirror image of a clock face reflects the entire dial left-right, which is equivalent to subtracting the shown time from 11 hours 60 minutes (i.e., from a base of 11:60).

- Mirror image of 3:15: $11:60 - 3:15 = 8:45$. So a clock reading 3:15 shows 8:45 in its mirror reflection.

4. Water Image of Clock Times (Formula-Based)

The water image of a clock face reflects the dial top-to-bottom, equivalent to subtracting the shown time from a base of 6:00 (adding 12 hours first if the result would be negative).

- Water image of 3:15: $6:00 - 3:15 = 2:45$. So a clock reading 3:15 shows 2:45 in its water reflection.

5. Mirror and Water Images of Numbers

For digits, only 0, 1, and 8 are symmetric enough to look unchanged under either type of flip; the safest way to reason about multi-digit numbers made purely of these digits is that a mirror image reverses their ORDER (since it's a left-right flip), while a water image keeps their order the same (since it's a top-bottom flip, not left-right).

- Mirror image of '1801' (all self-symmetric digits): reading order reverses to '1081'. Water image of '108' (all self-symmetric digits): order stays the same since only vertical position flips to '108'.

6. Symmetry Axis Identification

Before attempting a mirror or water image question on a letter or figure, identify whether it has a vertical line of symmetry (relevant for mirror images), a horizontal line of symmetry (relevant for water images), or neither — this quickly tells you whether the character will look unchanged, flipped, or entirely different.

- Letter 'H' has both a vertical and a horizontal line of symmetry, so it looks identical in both its mirror and water image. Letter 'F' has neither, so it looks different (and effectively unrecognizable as a normal letter) in both.

7. Combined Mirror + Water Image Problems

Some questions ask for the mirror image of an object's water image (or vice versa) — apply the two transformations strictly in the stated order, treating the output of the first transformation as the input to the second.

- Find the mirror image of the water image of 4:20. Step 1 (water image): $6:00 - 4:20 = 1:40$. Step 2 (mirror image of that result): $11:60 - 1:40 = 10:20$. Final answer: 10:20.

Common Pitfalls

- Confusing mirror image (left-right flip, reverses character ORDER) with water image (top-bottom flip, order stays the SAME).

- Using the mirror-time formula (11:60 " time) when the question actually asks for a water image, or vice versa.
- Assuming all digits/letters look unchanged under a flip, instead of checking each character's specific symmetry axis.
- In a combined mirror+water question, applying the two transformations in the wrong order.
- Forgetting to add 12 hours when a water-image time subtraction would otherwise produce a negative result.
- Assuming a character symmetric about one axis (say, vertical) is automatically also symmetric about the other (horizontal) axis.

Key Takeaways

- Mirror image = left-right flip: character order reverses, and each character's shape flips horizontally.
- Water image = top-bottom flip: character order stays the same, and each character's shape flips vertically.
- Mirror image of a clock time = 11:60 " given time. Water image of a clock time = 6:00 " given time (add 12h if negative).
- Only digits 0, 1, 8 and select letters (A, H, I, M, O, T, U, V, W, X, Y for vertical symmetry) look unchanged under a flip.
- For combined mirror+water problems, apply the transformations strictly in the order stated, using the first result as the next input.
- Always check a character's specific symmetry axis (vertical for mirror, horizontal for water) rather than assuming both.

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